

Stage-2 Hybrid TEC-AO Feedback Analysis

Fixed Model Failure, Improved Attended Run, Threshold Stability Statistics, and Control Strategy for Future Experiments

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1 Purpose of this report

This report merges the analysis of three related components of the Stage-2 spectrograph stabilisation work:

1. the general modelling argument comparing **adaptive warm-up-derived models** against a **fixed model**;
2. the **20 April fixed-model/fixed-prior run**, which initially stabilised but later failed under environmental/OPL drift;
3. the **22–24 April attended run**, which showed improved long-duration behaviour, recovery from a mid-run excursion, and clearer limitations of AO travel management.

The goal is to present the numerical evidence, figures, and control interpretation in one coherent document suitable for discussion with supervisors and later adaptation into a paper/report section.

Core message: the system sensitivities are not constant between experiments. A purely fixed model can become inaccurate or even wrong in sign. The most defensible strategy is therefore **adaptive warm-up modelling + TEC-primary control + AO fine trim**, with the fixed model used only as a weak prior.

2 Background: adaptive versus fixed modelling

The spectrograph response can be approximated locally using linear sensitivity models:

$$\Delta X_k = \alpha_T \Delta T_k + \alpha_P \Delta P_k + \alpha_H \Delta H_k + \eta_k, \quad (1)$$

$$\Delta Y_k = \beta_T \Delta T_k + \beta_P \Delta P_k + \beta_H \Delta H_k + \varepsilon_k, \quad (2)$$

$$\Delta \text{OPL}_k = g_T \Delta T_k + g_P \Delta P_k + g_H \Delta H_k + \xi_k. \quad (3)$$

Here ΔT_k , ΔP_k , and ΔH_k represent thermal, pressure, and humidity/water-vapour changes, while α , β , and g are sensitivity coefficients.

2.1 Fixed model assumption

A fixed model assumes that the coefficients

$$\alpha_T, \alpha_P, \alpha_H, \beta_T, \beta_P, \beta_H, g_T, g_P, g_H$$

remain valid across different days and different operating states. This would only be reasonable if the plant were close to time-invariant and the thermal/environmental coupling were repeatable.

Main problem: the experimental data show that the sensitivities change substantially between runs. In some cases even the **sign** changes, so a fixed controller can push in the wrong direction.

2.2 Adaptive warm-up model

Instead, the model coefficients can be estimated during the warm-up of each individual run:

$$\hat{\theta}_{\text{warmup}} = \arg \min_{\theta} \sum_{k \in \text{warmup}} (y_k - X_k \theta)^2 + \lambda \|\theta\|^2. \quad (4)$$

This gives run-specific coefficients for the current plant state.

Important point: the adaptive model does not assume that yesterday’s sensitivity is still valid today. It identifies the present response from the current run itself.

3 Earlier three-experiment comparison

The earlier comparison used three runs:

- **Experiment A:** `rt_stage2_hybrid_temp_ao_23_03_26.csv`
- **Experiment B:** `rt_stage2_final_v6.csv`
- **Experiment C:** `rt_stage2_20260415.fixed_model_hybrid_analysis_ready.csv`

3.1 Performance summary

Experiment	RMS dX (px)	RMS dY (px)	RMS drift (px)	Mean $ dY $ (px)	Frac($ dY \leq 0.10$)	OPL std (μm)
A	0.751	0.371	0.838	0.231	0.390	1.734
B	0.249	0.523	0.580	0.164	0.176	0.905
C	0.301	4.982	4.991	3.301	0.064	5.820

Table 1: Main comparison metrics across the earlier three experiments.

Highs:

- **Experiment A** gave the best centroid/image stability.
- **Experiment B** gave the calmest thermal/OPL behaviour.

Low:

- **Experiment C** was the worst overall, with very large dY , drift, OPL, and TEC variability.

3.2 Evidence that sensitivities change between runs

The strongest evidence comes from Experiment C, where the warm-up-fitted sensitivities differed strongly from the fixed model values.

Quantity	Warm-up fitted value	Fixed value used	Difference
g_T for OPL ($\mu\text{m}/^\circ\text{C}$)	-11.961	-4.100	-7.861
g_P for OPL ($\mu\text{m}/\text{hPa}$)	-6.773	+0.250	-7.023
g_H for OPL ($\mu\text{m}/(\text{g m}^{-3})$)	-0.390	-0.670	+0.280
β_T for dY (px/ $^\circ\text{C}$)	-4.033	+1.605	-5.638
β_P for dY (px/hPa)	-3.290	-0.633	-2.656
β_H for dY (px/(g m $^{-3}$))	-0.097	+0.742	-0.839

Table 2: Experiment C: warm-up-derived sensitivities compared with the fixed coefficients.

Critical result: these are not small calibration drifts. Several coefficients differ substantially and some change sign. If the sign is wrong, the controller may push in the wrong direction.

The comparison between warm-up fitted sensitivities and logged early adaptive/live coefficients also showed mismatch across experiments. For example:

- **Experiment A:** warm-up OPL temperature sensitivity $\approx +9.08 \mu\text{m}/^\circ\text{C}$, while logged early adaptive value $\approx +0.94 \mu\text{m}/^\circ\text{C}$.
- **Experiment B:** warm-up OPL temperature sensitivity $\approx +0.66 \mu\text{m}/^\circ\text{C}$, while logged early value $\approx -4.10 \mu\text{m}/^\circ\text{C}$.
- **Experiment C:** warm-up OPL temperature sensitivity $\approx -11.96 \mu\text{m}/^\circ\text{C}$, while logged early value $\approx -4.10 \mu\text{m}/^\circ\text{C}$.

Correct interpretation:

1. use warm-up identification to capture the current run;
2. use the adaptive result as the main model;
3. use the fixed model only as a weak prior;
4. keep the empirical image error as the main stabilising anchor.

4 Five-run A–E comparison using cleaned OPL logic

A broader comparison was then made across five experiments, labelled A–E:

- **A:** 23 March run, `rt_stage2_hybrid_temp_ao_23_03_26.csv`
- **B:** April v6 run, `rt_stage2_final_v6_recovered_full.csv`
- **C:** 15 April fixed-model run, `rt_stage2_20260415_fixed_model_hybrid_analysis_ready.csv`
- **D:** 20 April fixed-prior run, `rt_stage2_experimentD_v4_FIXED_analysis_ready.csv`
- **E:** 22–24 April attended run, `rt_stage2_experimentD_fresh_attended_analysis_ready.csv`

4.1 Date/file check

Run	Rows	Duration (h)	Start	End	Bad OPL rows
A_23Mar	1924	43.47	23 Mar 11:51	25 Mar 15:07	0
B_Apr_v6	1280	50.89	07 Apr 10:39	09 Apr 16:12	677
C_Apr15_fixed	1175	47.11	15 Apr 11:03	17 Apr 12:18	0
D_Apr20_fixed_prior	936	35.19	20 Apr 14:13	22 Apr 03:10	0
E_Apr22_attended	1238	45.24	22 Apr 09:10	24 Apr 10:48	0

Table 3: A–E file check and cleaned OPL status. B had many bad OPL rows, so cleaned OPL statistics are required for fair comparison.

4.2 Main A–E performance comparison

Run	RMS dX	RMS dY	RMS err	MAE err	Median err	P90 err	P95 err	Max err	≤ 0.5 px	≤ 1.0 px	TEC/hr	AO/hr
A_23Mar	0.7512	0.3711	0.8379	0.6612	0.4692	1.3695	1.4227	1.5770	51.56%	64.92%	5.11	2.39
B_Apr_v6	0.3266	1.7309	1.7615	1.0742	0.6585	2.2466	3.1387	9.4794	37.89%	70.78%	11.18	5.36
C_Apr15_fixed	0.3007	4.9816	4.9907	3.3427	1.9665	8.9788	11.7784	15.8802	23.32%	36.43%	8.62	0.70
D_Apr20_fixed_prior	0.4394	5.8756	5.8920	4.1951	2.0810	10.1990	10.3012	13.2485	28.95%	34.83%	13.19	1.08
E_Apr22_attended	0.5254	1.2015	1.3113	0.8726	0.5320	2.2155	3.1634	5.9438	47.82%	75.28%	5.79	2.41

Table 4: Main A–E comparison. RMS is included, but MAE, median, percentiles, and threshold fractions are also shown because they describe practical stability better than RMS alone.

4.3 Threshold stability comparison

Run	≤ 0.01	≤ 0.05	≤ 0.10	≤ 0.20	≤ 0.30	≤ 0.40	≤ 0.50	≤ 0.75	≤ 1.00
A_23Mar	0.05	2.23	13.15	28.22	38.36	44.70	51.56	58.99	64.92
B_Apr_v6	0.00	0.62	2.89	11.33	19.61	27.66	37.89	55.70	70.78
C_Apr15_fixed	0.09	0.68	3.15	8.94	12.09	15.83	23.32	31.49	36.43
D_Apr20_fixed_prior	0.00	2.99	9.83	17.74	24.47	27.24	28.95	31.94	34.83
E_Apr22_attended	0.16	3.39	8.24	20.68	30.86	39.01	47.82	62.52	75.28

Table 5: Percentage of samples within each radial-error threshold for A–E.

Run	≤ 0.01	≤ 0.05	≤ 0.10	≤ 0.20	≤ 0.30	≤ 0.40	≤ 0.50	≤ 0.75	≤ 1.00
A_23Mar	0.02	0.98	5.77	12.87	17.76	20.66	23.88	27.00	31.54
B_Apr_v6	0.00	0.42	1.58	6.74	10.97	15.08	20.30	29.41	37.11
C_Apr15_fixed	0.04	0.32	1.48	4.19	5.67	7.57	11.15	15.37	17.68
D_Apr20_fixed_prior	0.00	1.06	3.50	6.31	8.69	9.67	10.26	11.32	12.33
E_Apr22_attended	0.07	1.54	3.75	9.40	14.03	17.74	21.73	28.40	34.18

Table 6: Total hours within each radial-error threshold for A–E.

4.4 Simple ranking

Run	RMS err	MAE err	Median err	P90 err	≤ 0.5 px	Overall score
A_23Mar	0.8379	0.6612	0.4692	1.3695	51.56%	7
E_Apr22_attended	1.3113	0.8726	0.5320	2.2155	47.82%	9
B_Apr_v6	1.7615	1.0742	0.6585	2.2466	37.89%	14
C_Apr15_fixed	4.9907	3.3427	1.9665	8.9788	23.32%	21
D_Apr20_fixed_prior	5.8920	4.1951	2.0810	10.1990	28.95%	24

Table 7: Simple ranking using multiple practical stability metrics. Lower overall score is better.

A–E conclusion: A remains the best by RMS/median stability, E is the strongest recent long-duration recovery run, B has good OPL/thermal calmness but weaker pixel stability, while C and D are the clearest fixed/fixed-prior failure cases.

5 20 April fixed-model/fixed-prior experiment

5.1 Objective

The 20 April experiment evaluated a fixed-model TEC feedback controller applied to spectrograph stabilisation. The controller used pre-derived coefficients to predict and correct optical path length and centroid drift. In this report it is described as a **fixed-prior / fixed-model-guided run**, because the controller was guided by previously derived sensitivity values rather than relying fully on a newly identified adaptive warm-up model for that specific run.

The key questions were:

- Can the fixed model stabilise dY ?
- Does TEC maintain authority under environmental and OPL drift?
- Does AO remain a fine stabiliser, or does it become involved in large-drift behaviour?

- Does the fixed-prior strategy remain valid over long duration?

5.2 Was the 20 April run really using fixed values?

The 20 April controller should be described carefully as a **fixed-prior / fixed-model-guided run**, not a fully adaptive warm-up model run. The fixed coefficients used as priors were:

$$\text{FIXED_BETA_Y} = [1.6050, -0.6333, 0.7422], \quad (5)$$

$$\text{FIXED_G}_T = -4.10 \mu\text{m}/^\circ\text{C}, \quad (6)$$

$$\text{FIXED_G}_P = +0.25 \mu\text{m}/\text{hPa}, \quad (7)$$

$$\text{FIXED_G}_{\text{H}_2\text{O}} = -0.67 \mu\text{m}/(\text{g m}^{-3}). \quad (8)$$

The run did include a warm-up/initialisation period, but the warm-up was not used as the dominant adaptive model in the way now recommended. Instead, the previous-run coefficients remained central as fixed priors or reference values.

Meeting wording: this was not a pure no-model run and not a fully adaptive warm-up model run. It was a fixed-prior guided run, which worked while the plant remained close to the assumed sensitivity, but failed when the current run drifted away from that assumption.

5.3 Dataset and overall metrics

Metric	Value
Duration	36.95 h
Samples	936
RMS dX	0.4394 px
RMS dY	5.8756 px
RMS radial error	5.8920 px
Mean absolute error	4.1951 px
Median radial error	2.0810 px
Median $ dY $	2.0666 px
90th percentile $ dY $	10.1775 px
95th percentile $ dY $	10.2833 px
Maximum $ dY $	13.2478 px
TEC actions	464
AO actions	38
TEC actions per hour	13.19 h ⁻¹
AO actions per hour	1.08 h ⁻¹
OPL span	19.1840 μm
OPL standard deviation	4.7695 μm

Table 8: Overall performance of the 20 April fixed-prior run.

5.4 Full system behaviour

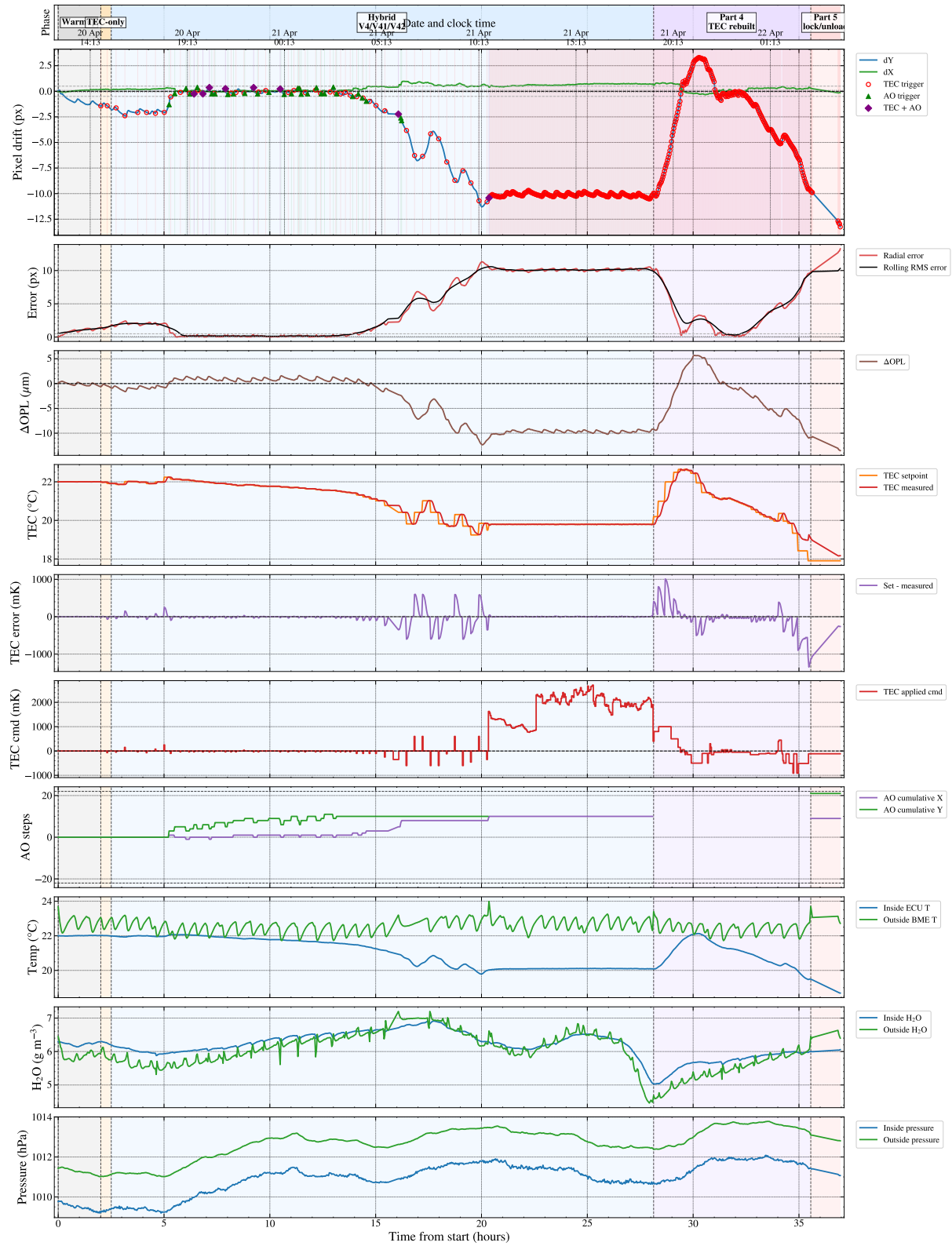


Figure 1: 20 April fixed-prior run: full time-series showing pixel drift, radial error, OPL variation, TEC control, AO response, and environmental parameters. Highlighted regions show behavioural phases.

5.5 Behavioural windows

The run showed four main regimes:

1. stable hybrid plateau;
2. breakaway;
3. bad negative plateau;
4. failed recovery.

Window	Time range (h)	RMS dY (px)	RMS err (px)	Mean dY (px)	OPL change (μm)	TEC/AO actions
Stable plateau	5.01–14.48	0.3801	0.3949	-0.1447	+0.8558	30 / 33
Breakaway	14.52–20.47	6.6485	6.6793	-5.8704	-10.1123	19 / 5
Bad negative plateau	20.51–28.08	10.1119	10.1328	-10.1107	+1.0174	200 / 0
Failed recovery	28.13–36.95	4.9288	4.9434	-2.8499	-4.3133	208 / 0

Table 9: Event-window analysis of the 20 April fixed-prior run.

Positive result: the fixed model initially produced a stable plateau with RMS $dY = 0.3801$ px and RMS radial error = 0.3949 px. This is strong evidence that the fixed model can work temporarily when the system remains close to the assumed operating state.

Failure mode: after the breakaway, the system entered a sustained negative plateau near $dY \approx -10$ px. TEC continued to act but did not recover the centroid.

5.6 Threshold stability statistics for 20 April

RMS alone does not show how often the controller reached very small errors. Therefore cumulative threshold statistics were also computed.

Region	≤ 0.01 px	≤ 0.05 px	≤ 0.10 px	≤ 0.20 px	≤ 0.50 px	≤ 1.00 px
Full run, samples	0	28	92	166	271	326
Full run, hours	0.00	1.06	3.50	6.31	10.26	12.33
Full run, percent	0.00	2.99	9.83	17.74	28.95	34.83
Best plateau, samples	0	27	87	154	226	242
Best plateau, hours	0.00	1.03	3.31	5.86	8.60	9.21
Best plateau, percent	0.00	10.84	34.94	61.85	90.76	97.19

Table 10: 20 April threshold stability statistics using radial error. The fixed-prior model achieved excellent short-term stability during the best plateau, but not over the full run.

Important nuance: the 20 April fixed-prior run was not simply bad from the beginning. Its best plateau spent **90.76%** of samples within **0.5 px** and **97.19%** within **1.0 px**. The failure was a long-duration robustness failure, not a failure to ever stabilise.

5.7 TEC command to pixel-response lag

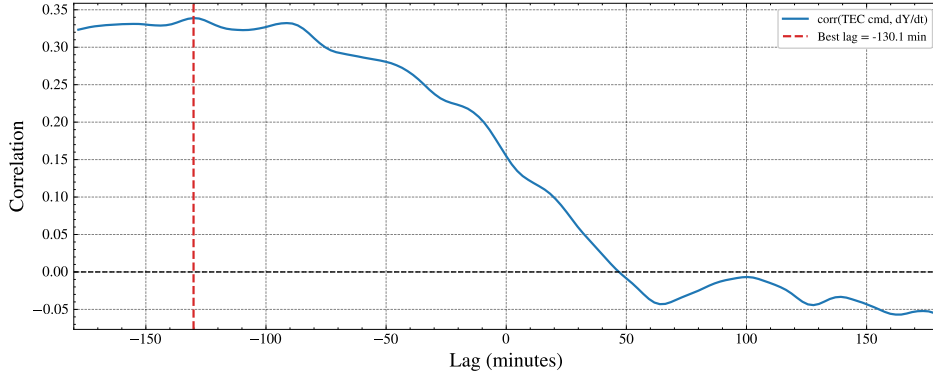


Figure 2: 20 April run: lag correlation between TEC command and pixel-response derivative.

The lag analysis gave:

- best lag: **-130.1 min**;
- correlation at best lag: **0.339**.

Interpretation: the negative lag suggests that pixel drift was already changing before the TEC correction. The controller was reacting to an environmental/OPL-driven drift rather than predicting it.

5.8 Likely trigger of the 20 April drift

From the available evidence, the 20 April drift is not best explained as a simple TEC setpoint tracking error. The strongest clues are:

- the breakaway window had a large OPL change of approximately **-10.11 μm** ;
- the best TEC–pixel lag was **-130.1 min**, meaning the pixel motion appears to lead the TEC response;
- the fixed-prior model had previously assumed coefficients that may no longer matched the current plant;
- pressure and water-vapour changes probably contributed through refractive-index/OPL coupling, but the current evidence does not isolate one single environmental variable as the sole trigger.

Most defensible conclusion: the April 20 run drifted because the fixed-prior model lost validity as the internal thermal/OPL/environmental state changed. TEC then chased the drift and settled near a wrong equilibrium around $dY \approx -10$ px.

6 22–24 April attended hybrid experiment

6.1 Overview

The 22–24 April attended run tested an improved TEC-primary controller with AO-assisted stabilisation. The experiment ran over approximately 50 hours and included intentional restart/recovery phases. This run is important because it did not merely show a short stable interval; it also showed recovery from a large mid-run excursion and avoided the sustained -10 to -13 px runaway seen in the 20 April run.

6.2 Dataset summary

Metric	Value
Start time	22 Apr 2026, 09:10
End time	24 Apr 2026, 10:48
Total duration	49.63 h
Valid science rows	1238
TEC actions	262
AO actions	109
TEC actions per hour	5.79 h ⁻¹
AO actions per hour	2.41 h ⁻¹
OPL span	9.9780 μm
OPL standard deviation	1.9085 μm

Table 11: Dataset summary for the 22–24 April attended run.

6.3 Full-run performance

Metric	Value
RMS dX	0.5254 px
RMS dY	1.2015 px
RMS radial error	1.3113 px
Mean absolute error	0.8726 px
Median radial error	0.5320 px
Median $ dY $	0.3989 px
90th percentile $ dY $	2.0604 px
95th percentile $ dY $	2.8079 px
Maximum $ dY $	5.6709 px
Fraction $ err \leq 0.1$ px	8.24%
Fraction $ err \leq 0.2$ px	20.68%
Fraction $ err \leq 0.5$ px	47.82%
Fraction $ err \leq 1.0$ px	75.28%

Table 12: Full-run performance metrics for the 22–24 April attended experiment.

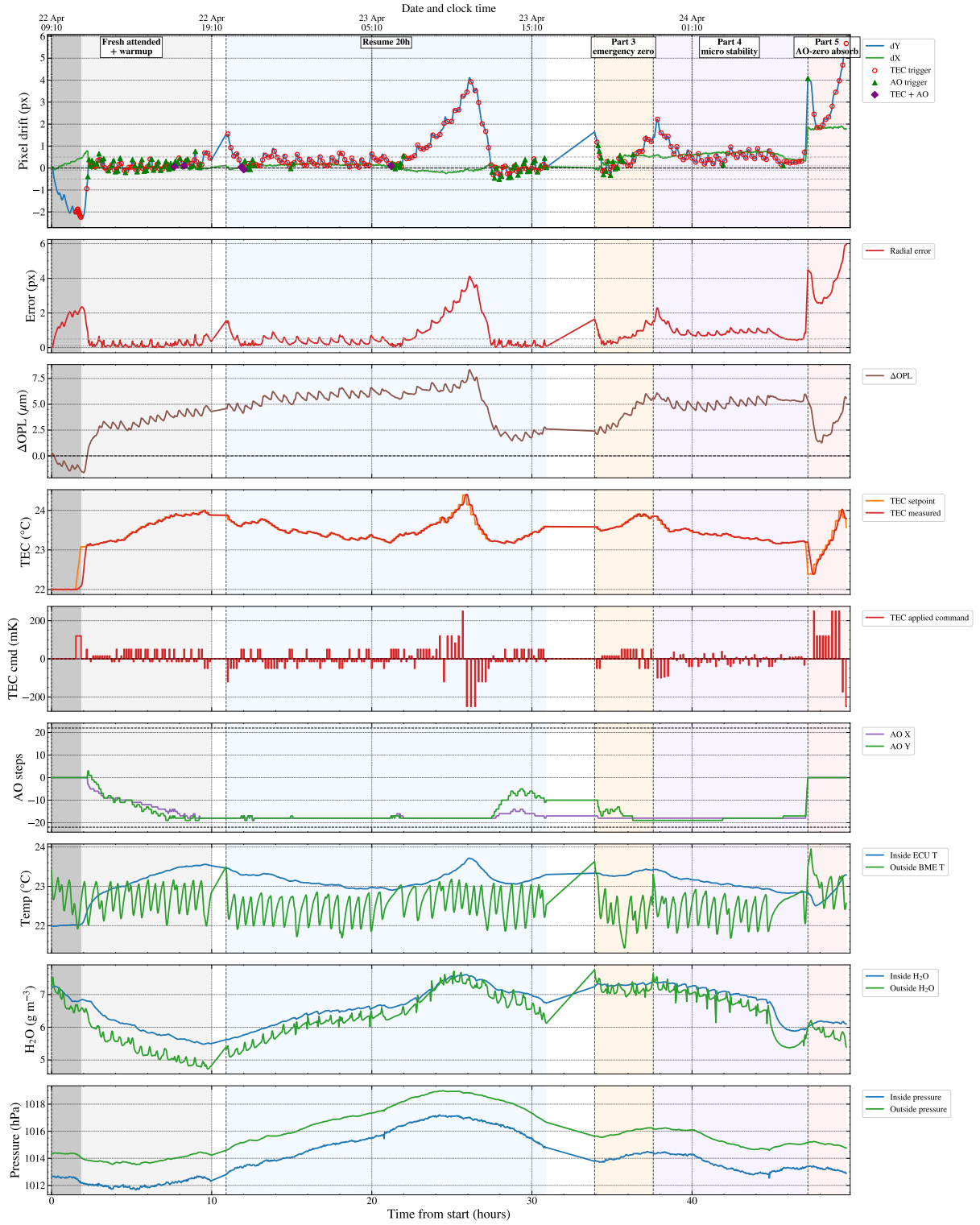


Figure 3: 22–24 April attended run: full multi-panel behaviour showing pixel drift, radial error, OPL, TEC setpoint/measured temperature, TEC commands, AO cumulative steps, and environmental channels.

6.4 Performance excluding final AO-zero absorb section

The final Part 5 AO-zero/TEC absorb section was a deliberate recentering/absorb experiment and should not be treated as nominal stability data.

Metric	Value
Duration	47.10 h
RMS dX	0.3379 px
RMS dY	0.9778 px
RMS radial error	1.0345 px
Fraction $ err \leq 0.1$ px	8.67%
Fraction $ err \leq 0.2$ px	21.77%
Fraction $ err \leq 0.5$ px	50.34%

Table 13: 22–24 April performance excluding the deliberate final AO-zero absorb/recentering section.

6.5 Phase-based performance

Phase	Duration (h)	RMS dX (px)	RMS dY (px)	RMS err (px)
Fresh attended	9.97	0.1778	0.8131	0.8323
Resume 20h	20.00	0.1322	1.1826	1.1900
Part 3 emergency zero	3.60	0.4294	0.7178	0.8364
Part 4 micro-stability	9.52	0.6159	0.7078	0.9382
Part 5 AO-zero absorb	2.40	1.8294	3.2696	3.7466

Table 14: RunLabel-based phase performance in the 22–24 April attended run.

6.6 Trimmed feedback-zone performance

A trimmed presentation dataset retained the useful warm-up/stable-feedback/recovered-feedback/micro-stability regions while excluding the large deliberate excursions.

Metric	Value
Rows	886
Clock coverage	22 Apr 09:10 – 24 Apr 08:15
RMS dX	0.3569 px
RMS dY	0.6467 px
RMS radial error	0.7386 px
Median $ dY $	0.3325 px
90th percentile $ dY $	0.9479 px
95th percentile $ dY $	1.5340 px
Maximum $ dY $	2.3157 px
Fraction $ err \leq 0.1$ px	9.48%
Fraction $ err \leq 0.2$ px	24.60%
Fraction $ err \leq 0.5$ px	56.66%
Fraction $ err \leq 1.0$ px	86.79%

Table 15: Trimmed feedback-zone performance for the 22–24 April attended run.

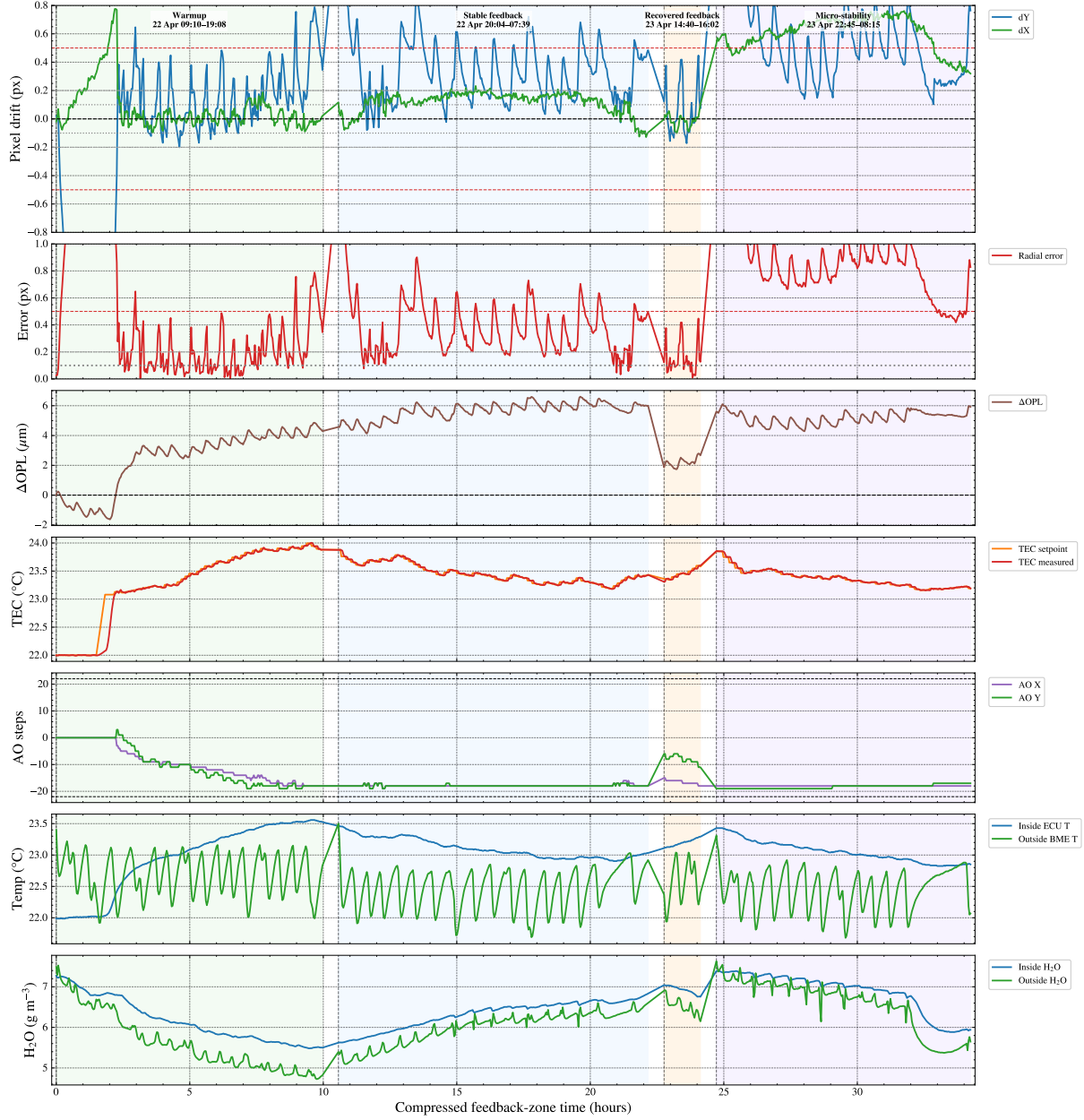


Figure 4: 22–24 April: trimmed feedback-zone plot focusing on useful control regions and sub-pixel behaviour.

Practical reading: the full run gives the honest overall number, but the trimmed feedback-zone result better represents useful controller performance. In the trimmed control regions, RMS dY was 0.6467 px and 56.66% of points were within 0.5 px radial error.

7 Triangle excursion in the 22–24 April run

7.1 Observation

A smooth triangular dY excursion occurred during the Resume 20h section. The event occurred approximately over:

$$22.5 \leq t_{\text{hr}} \leq 29.5,$$

corresponding approximately to 23 Apr 07:42–14:38. The dY value rose to a peak near 4 px and then recovered close to zero by approximately 15:10 on 23 April.

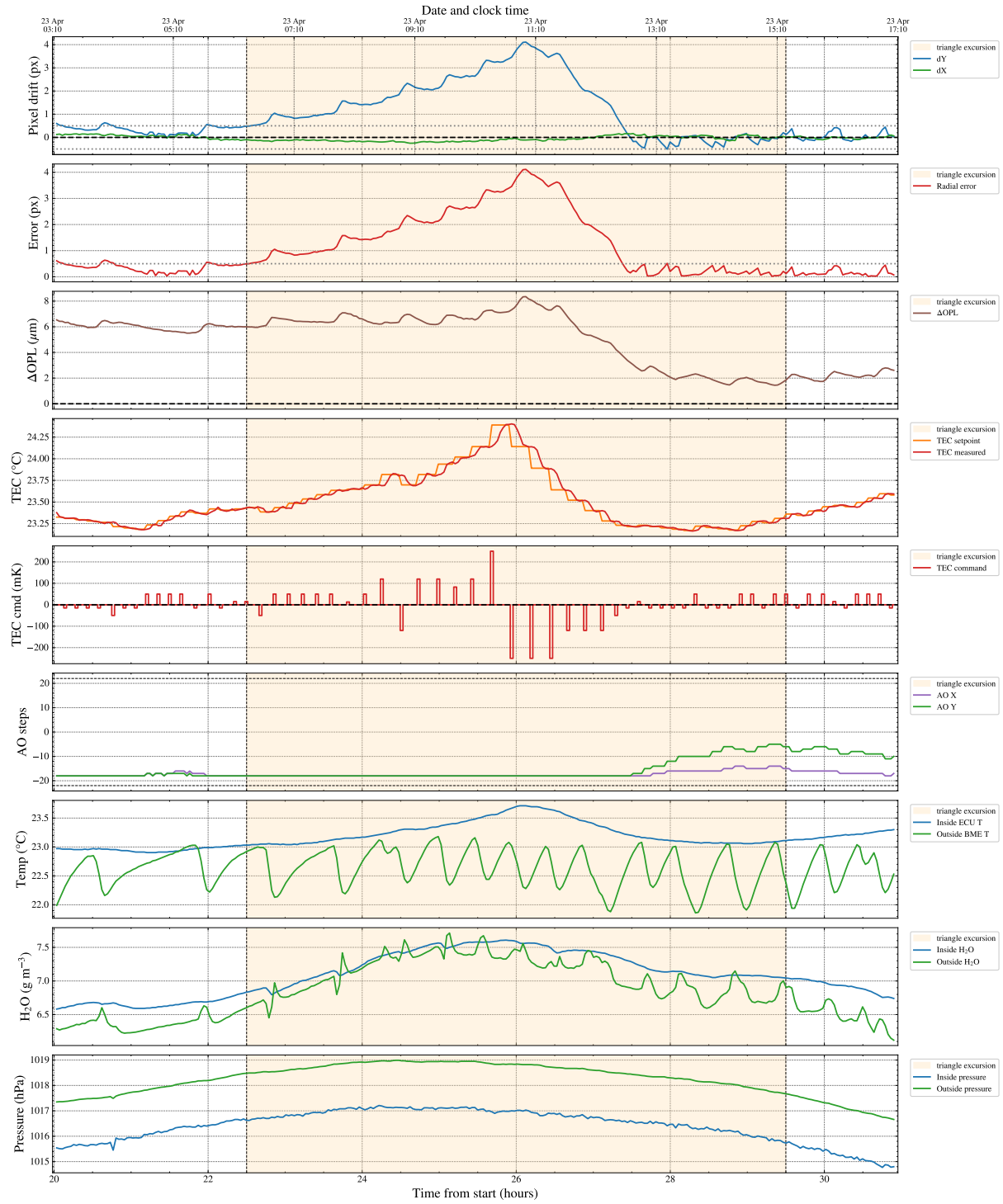


Figure 5: 22–24 April: zoomed diagnostic plot of the triangular dY excursion.

7.2 Window statistics

Window	Duration (h)	RMS dX (px)	RMS dY (px)	RMS err (px)	AO motion
Pre-triangle stable control	11.60	0.1405	0.4331	0.4553	$(-18, -18) \rightarrow (-18, -18)$
Triangle excursion	6.93	0.1295	1.9232	1.9276	$(-18, -18) \rightarrow (-15, -6)$
Post-triangle recovery	1.40	0.0479	0.1867	0.1927	$(-15, -6) \rightarrow (-17, -10)$
Micro-stability Part 4	9.52	0.6159	0.7078	0.9382	$(-18, -19) \rightarrow (-18, -17)$

Table 16: Selected event-window performance for the 22–24 April attended run.

7.3 Correlation analysis during the triangle

During the triangular excursion, dY correlated most strongly with internal temperature, TEC measured temperature, TEC setpoint, OPL, humidity, and pressure.

Variable	Correlation with dY
ECU/internal temperature	0.919
TEC measured temperature	0.913
TEC setpoint	0.865
Δ OPL	0.848
Inside H ₂ O	0.802
Outside H ₂ O	0.789
Outside pressure	0.742
Inside pressure	0.706
AO cumulative Y	-0.645
AO cumulative X	-0.604
TEC command	-0.112
Outside BME temperature	0.094

Table 17: Correlation with dY during the triangular excursion.

Interpretation: the event was not primarily a direct outside-temperature event. It was most strongly coupled to internal temperature, TEC temperature, OPL, and water-vapour/pressure changes.

7.4 Rise and recovery mechanism

During the triangle rise:

Quantity	Start	Peak	Change
dY	0.5344 px	4.1104 px	+3.5759 px
dX	-0.0918 px	-0.1128 px	-0.0210 px
Radial error	0.5423 px	4.1119 px	+3.5697 px
ΔOPL	6.2384 μm	8.3477 μm	+2.1093 μm
TEC setpoint	23.4200°C	24.1406°C	+0.7206°C
TEC measured	23.3739°C	24.1527°C	+0.7787°C
ECU temperature	22.9890°C	23.7130°C	+0.7240°C
ECU H_2O	6.6954 g m ⁻³	7.5621 g m ⁻³	+0.8667 g m ⁻³
ECU pressure	1016.43 hPa	1017.03 hPa	+0.60 hPa
BME temperature	22.22°C	22.28°C	+0.06°C
BME H_2O	6.4770 g m ⁻³	7.4499 g m ⁻³	+0.9729 g m ⁻³
BME pressure	1018.20 hPa	1018.82 hPa	+0.62 hPa
AO cumulative position	(-18, -18)	(-18, -18)	no motion
TEC actions		20	net +0.77065°C

Table 18: Triangle rise: changes from start to dY peak.

During the recovery/fall:

Quantity	Start	End	Change
dY	4.0249 px	0.0308 px	-3.9941 px
dX	-0.1069 px	0.0644 px	+0.1714 px
Radial error	4.0263 px	0.0714 px	-3.9549 px
ΔOPL	8.1672 μm	2.6087 μm	-5.5584 μm
TEC setpoint	24.1406°C	23.5806°C	-0.5600°C
TEC measured	24.1371°C	23.5933°C	-0.5438°C
ECU temperature	23.7050°C	23.3000°C	-0.4050°C
ECU H_2O	7.5648 g m ⁻³	6.7400 g m ⁻³	-0.8248 g m ⁻³
ECU pressure	1017.00 hPa	1014.80 hPa	-2.20 hPa
BME H_2O	7.3315 g m ⁻³	6.1194 g m ⁻³	-1.2121 g m ⁻³
BME pressure	1018.82 hPa	1016.66 hPa	-2.16 hPa
AO cumulative position	(-18, -18)	(-17, -10)	AO Y unloaded by +8 steps
TEC actions		30	net -0.56°C

Table 19: Triangle recovery/fall: changes from peak region to recovery.

7.5 Physical explanation of the triangle

The triangle was not caused by AO motion during the rise, since AO remained fixed at (-18, -18). Instead, AO was blocked or unavailable, and the controller relied on TEC.

Layman explanation: TEC is the slow but powerful steering wheel; AO is the fast but limited fine knob. During the triangle rise, the fine knob was already near its end stop, so the controller used the big steering wheel. The TEC temperature increase moved the optical path and shifted the spectrum in Y . Later, when the controller reversed and AO regained some usable range, the system returned near zero.

In more technical terms:

- AO was near its negative travel boundary and did not move during the rise.
- The controller applied a net positive TEC response of approximately +0.77°C.
- TEC/internal temperature increased, OPL increased, and dY increased to approximately 4.11 px.

- Once TEC reversed and AO began unloading, dY returned close to zero by about 15:10 on 23 April.

Control issue identified: if AO is saturated, TEC must not be allowed to make a large recentering move without predictive damping and sign/authority checks. Otherwise, the TEC can create a controlled but large thermal excursion.

8 RMS versus threshold statistics

8.1 Why RMS can look worse than the console impression

During the experiment, the console sometimes showed dX and dY values close to 0.01 px. This is real and important. However, RMS is calculated over an entire selected region:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N e_i^2}. \quad (9)$$

Because the error is squared, a small number of large excursions can dominate the RMS. Therefore:

- RMS tells us about overall stability including excursions;
- threshold statistics tell us how often the controller reached a given precision;
- median and percentile values show the typical behaviour more fairly than RMS alone;
- all of these metrics are needed for a fair evaluation.

Meeting explanation: RMS is not saying the controller never reached 0.01 px. It is saying that it did not hold that level continuously, and large excursions strongly penalised the overall number.

8.2 Cumulative threshold/histogram comparison

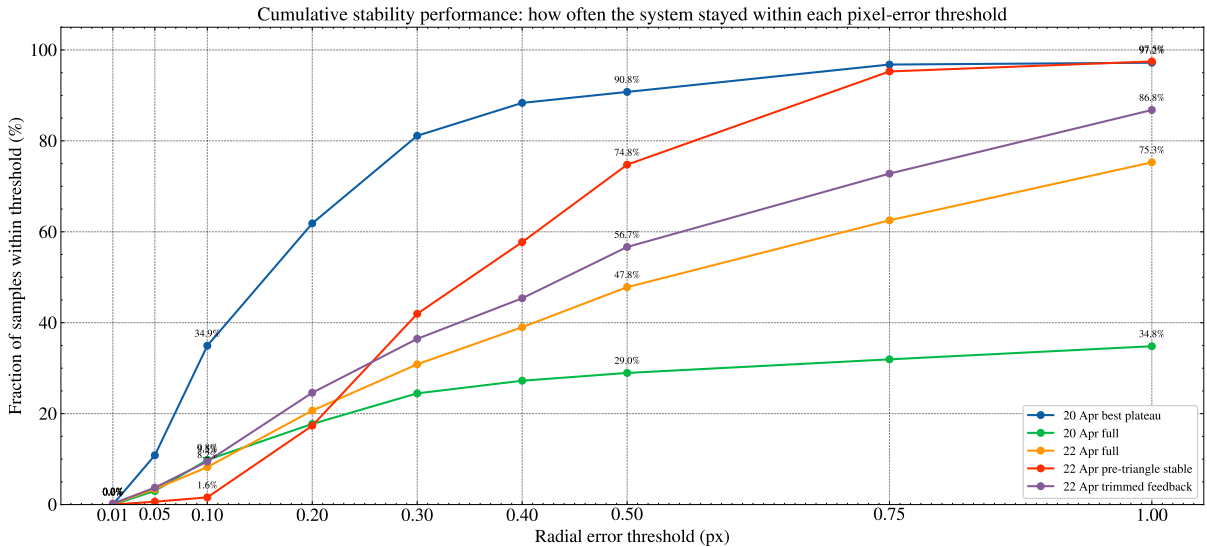


Figure 6: Cumulative threshold performance: fraction of samples within each radial-error threshold for the April 20 and April 22–24 runs.

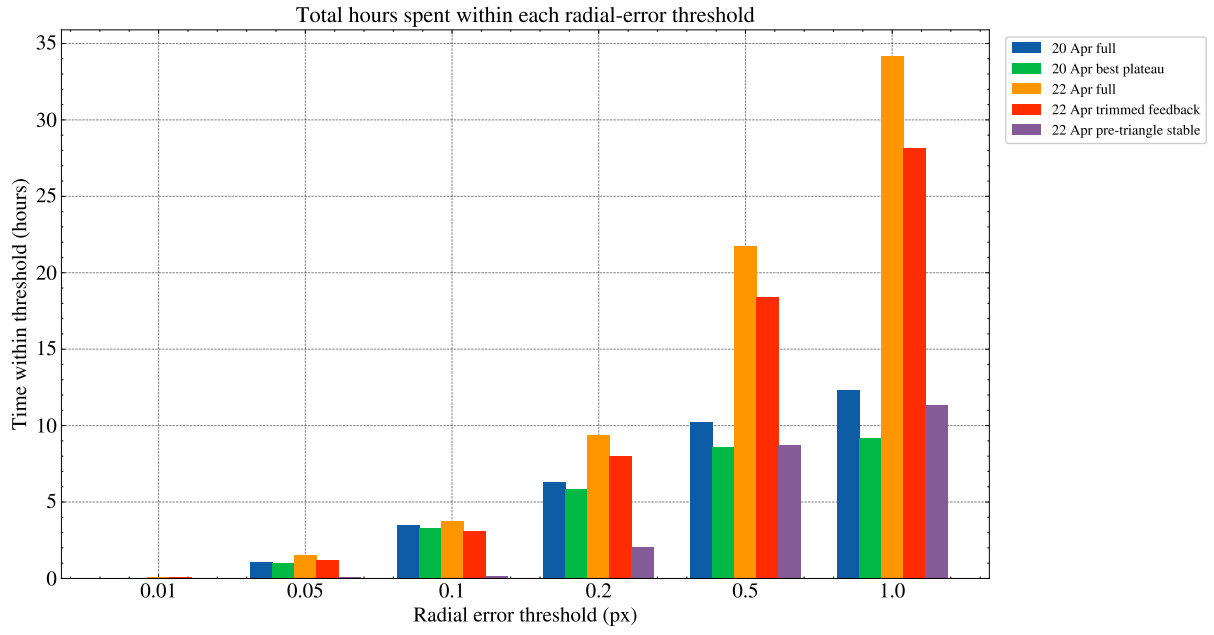


Figure 7: Total hours spent within selected radial-error thresholds for the April 20 and April 22–24 runs.

8.3 Detailed threshold table

Region	≤ 0.01 px	≤ 0.05 px	≤ 0.10 px	≤ 0.20 px	≤ 0.50 px	≤ 1.00 px
20 Apr full, samples	0	28	92	166	271	326
20 Apr full, hours	0.00	1.06	3.50	6.31	10.26	12.33
20 Apr full, percent	0.00	2.99	9.83	17.74	28.95	34.83
20 Apr best plateau, samples	0	27	87	154	226	242
20 Apr best plateau, hours	0.00	1.03	3.31	5.86	8.60	9.21
20 Apr best plateau, percent	0.00	10.84	34.94	61.85	90.76	97.19
22 Apr full, samples	2	42	102	256	592	932
22 Apr full, hours	0.07	1.54	3.75	9.40	21.73	34.18
22 Apr full, percent	0.16	3.39	8.24	20.68	47.82	75.28
22 Apr trimmed feedback, samples	2	33	84	218	502	769
22 Apr trimmed feedback, hours	0.07	1.21	3.09	8.00	18.42	28.19
22 Apr trimmed feedback, percent	0.23	3.72	9.48	24.60	56.66	86.79
22 Apr pre-triangle stable, samples	0	2	5	55	237	309
22 Apr pre-triangle stable, hours	0.00	0.07	0.19	2.02	8.70	11.34
22 Apr pre-triangle stable, percent	0.00	0.63	1.58	17.35	74.76	97.48

Table 20: Threshold statistics for radial error. This table shows both precision frequency and useful stable time.

Key comparison: the 20 April best plateau was the tightest short window, reaching **90.76%** within 0.5 px. However, the 22 April run produced more useful stable time over a much longer experiment, including **21.73 h within 0.5 px** over the full run and **18.42 h within 0.5 px** in the trimmed feedback zone.

9 Direct comparison: 20 April versus 22–24 April

Metric	20 April fixed-prior run	22–24 April attended run
Duration	36.95 h	49.63 h
Samples	936	1238
RMS dY full run	5.8756 px	1.2015 px
RMS radial error full run	5.8920 px	1.3113 px
Mean absolute error	4.1951 px	0.8726 px
Median radial error	2.0810 px	0.5320 px
Median $ dY $	2.0666 px	0.3989 px
90th percentile $ dY $	10.1775 px	2.0604 px
95th percentile $ dY $	10.2833 px	2.8079 px
Maximum $ dY $	13.2478 px	5.6709 px
Fraction $ err \leq 0.1$ px	9.83%	8.24%
Fraction $ err \leq 0.2$ px	17.74%	20.68%
Fraction $ err \leq 0.5$ px	28.95%	47.82%
Fraction $ err \leq 1.0$ px	34.83%	75.28%
Hours $ err \leq 0.5$ px	10.26 h	21.73 h
Hours $ err \leq 1.0$ px	12.33 h	34.18 h
TEC actions	464	262
AO actions	38	109
Worst behaviour	sustained -10 to -13 px runaway	recovered triangle + final deliberate AO reset
Nominal/control RMS dY	0.3801 px initially; later failed	0.6467 px trimmed feedback-zone
Recovery quality	poor; failed recovery	recovered near zero after excursion

Table 21: Direct comparison between the 20 April fixed-prior run and the 22–24 April attended run.

9.1 Percentage improvement from 20 April to 22–24 April

Using the full-run metrics, the 22–24 April attended run improved substantially relative to the 20 April fixed-prior run:

Metric	20 Apr	22–24 Apr	Improvement
RMS dY	5.8756 px	1.2015 px	79.6% reduction
RMS radial error	5.8920 px	1.3113 px	77.7% reduction
Mean absolute error	4.1951 px	0.8726 px	79.2% reduction
Median radial error	2.0810 px	0.5320 px	74.4% reduction
Median $ dY $	2.0666 px	0.3989 px	80.7% reduction
95th percentile $ dY $	10.2833 px	2.8079 px	72.7% reduction
Time within 0.5 px	10.26 h	21.73 h	2.12 $\times$ longer
Time within 1.0 px	12.33 h	34.18 h	2.77 $\times$ longer
TEC actions	464	262	43.5% fewer actions

Table 22: Main quantitative improvement from the 20 April run to the 22–24 April attended run.

Main improvement: the 22–24 April run avoided the sustained -10 to -13 px runaway seen on 20 April and recovered from the main triangular excursion. Full-run RMS dY improved by approximately **79.6%** and RMS radial error improved by approximately **77.7%**.

Remaining limitation: the 22–24 April run still showed ripple, AO travel-limit issues, and a TEC-driven triangular excursion when AO was saturated.

9.2 Which run is best?

The answer depends on what metric is used:

- **Best short stable window:** 20 April best plateau, RMS radial error = 0.3921 px and 90.76% within 0.5 px.
- **Best long-duration robustness among the two April runs:** 22–24 April attended run, because it avoided the sustained runaway and gave 21.73 h within 0.5 px over the full run.
- **Best useful control-region result:** 22–24 April trimmed feedback zone, RMS radial error = 0.7386 px and 56.66% within 0.5 px.
- **Best overall among A–E by ranking:** A_23Mar by RMS/median error, followed by E_Apr22_attended.

Final comparison statement: the fixed-prior model can achieve excellent short-term stability, but it is not robust over long duration. The 22–24 April attended/adaptive-style run is noisier than the best short plateau, but it is safer and more useful because it recovers and avoids the large sustained runaway.

10 Control strategy implications

10.1 Empirical correction

The empirical temperature correction from measured image drift is:

$$e_T^{(\text{emp})} = -\frac{\Delta Y}{\hat{\beta}_{Y,T}}. \quad (10)$$

This is useful because it directly ties the temperature correction to the measured image motion.

10.2 Model-based correction

A model-based correction can be estimated using both pixel and OPL information:

$$e_T^{(\text{px})} = f_{\text{pixel}}^{-1}(\Delta X, \Delta Y, \Delta P, \Delta H), \quad (11)$$

$$e_T^{(\text{opl})} = -\frac{\Delta \text{OPL}}{\hat{g}_T}. \quad (12)$$

These can be combined as:

$$e_T^{(\text{model})} = \frac{1}{2} \left(e_T^{(\text{px})} + e_T^{(\text{opl})} \right). \quad (13)$$

10.3 Confidence-weighted fusion

The final correction estimate may be written as:

$$e_T^{(\text{fused})} = (1 - w_m - w_f)e_T^{(\text{emp})} + w_me_T^{(\text{model})} + w_fe_T^{(\text{fixed})}, \quad (14)$$

where w_m depends on model confidence and w_f is small.

Meaning: the controller should not blindly trust the model. It should remain anchored to measured image error, while using OPL/environment models as supporting information.

10.4 TEC control law

A conservative event-based TEC control law can be written as:

$$\Delta u_k = \text{sat} \left(K_P e_k + K_I \sum_{j=1}^k e_j + K_D \dot{e}_k \right). \quad (15)$$

The controller should include:

- step caps;
- holdoff/settle windows;
- reversal freeze;
- persistence requirement;
- calm gating;
- predictive damping during AO-saturated states.

10.5 AO role

AO should not be the primary stabiliser. Its role should be:

- fine trim only;
- only when TEC is calm;
- only when residual error is modest;
- with active monitoring of cumulative travel.

The AO step estimate is:

$$\Delta s = M^{-1} \begin{bmatrix} -\Delta X \\ -\Delta Y \end{bmatrix}, \quad (16)$$

where M is the AO calibration matrix.

Correct AO philosophy: AO should trim the residual, not rescue a badly unstable thermal loop.

11 Advantages and limitations of regression-based adaptive modelling

11.1 Advantages

- simple and interpretable;
- fast to compute;
- suitable for warm-up data windows;
- physically understandable;
- easy to explain in supervisor discussions and reports.

11.2 Limitations

- assumes approximate local linearity;
- can be affected by collinearity;
- may overfit if warm-up data are poor;
- does not naturally capture lag, hysteresis, or hidden thermal state;
- cannot by itself solve AO travel-limit saturation.

Practical conclusion:

adaptive warm-up regression + confidence gating + TEC-primary control + AO fine trim

12 Recommended next-run logic

The next controller should keep the strongest lessons from both experiments:

1. Use a **warm-up identification stage** to estimate current-run sensitivities.
2. Use a **TEC-only settling stage** before enabling AO.
3. Keep **TEC as the primary actuator** for OPL/Y drift.
4. Use **AO only as fine trim**, not as the main rescue actuator.
5. Add **proactive AO unload/recentering** before AO reaches its physical travel boundary.
6. Reduce TEC authority or add stronger damping when AO is near saturation.
7. Use explicit environmental predictors such as H₂O and pressure rather than raw RH alone.
8. Report both RMS and threshold statistics in future performance summaries.
9. Track median, MAE, P90/P95, and useful hours within error thresholds, not only RMS.

One-line final strategy:

B-style calm TEC backbone + adaptive current-run model + A-style limited AO trim + AO travel manager

13 Final conclusion

The experiments together show a clear progression.

The earlier A/B/C comparison established that fixed coefficients are not reliable across runs. Experiment A showed that good image stability is achievable; Experiment B showed that calmer OPL/thermal behaviour is possible; Experiment C showed that mismatched fixed coefficients and unstable control logic can drive large drift.

The A–E comparison confirms this pattern numerically. A_23Mar remains the best overall by RMS radial error, MAE, median error, and time within 0.5 px. E_Apr22_attended is the strongest recent long-duration recovery run and ranks second overall, with 75.28% of samples within 1.0 px and 34.18 h within 1.0 px. C_Apr15_fixed and D_Apr20_fixed_prior are the clearest examples of fixed/fixed-prior strategies becoming unstable over long duration.

The 20 April run demonstrated the fixed-prior failure mode directly in the Stage-2 hybrid system. It initially reached a stable plateau with RMS $dY = 0.3801$ px and 90.76% of the best-plateau samples within 0.5 px radial error. However, it later entered a large breakaway and bad negative plateau near $dY \approx -10$ px. Lag analysis indicated that the controller was reacting to drift that had already begun.

The 22–24 April attended run showed a major improvement. Full-run RMS dY fell to 1.2015 px, corresponding to a 79.6% reduction relative to the 20 April run. The trimmed useful feedback-zone RMS dY was 0.6467 px, with 56.66% of points within 0.5 px and 86.79% within 1.0 px. The system did not suffer the sustained -10 to -13 px runaway seen on 20 April. It also recovered from a triangular mid-run excursion. However, that event revealed a remaining limitation: when AO was saturated, TEC applied a large thermal move, producing a temporary OPL and dY excursion.

Final engineering conclusion: the next controller should use an adaptive warm-up-derived model, TEC as the primary actuator, and AO only as fine trim. It must also manage AO travel proactively and damp TEC moves when AO is saturated.